Linear Algebra II HW 9

1. Find an orthogonal or unitary diagonalizing matrix for each of the following:

(a) $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$

(b) $\begin{pmatrix} 1 & 3+i \\ 3-i & 4 \end{pmatrix}$

(c) $\begin{pmatrix} 2 & i & 0 \\ -i & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$

Solution

There will not be a unique unitary diagonalizing matrix for a given Hermitian matrix A, however, the column vectors of any unitary diagonalizing matrix must be unit eigenvectors of A.

(a) $\lambda_1 = 3$ has a unit eigenvector $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right)^T$

$\lambda_2 = 1$ has a unit eigenvector $\left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \right)^T$.

$$Q = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

(b) $\lambda_1 = 6$ has a unit eigenvector $\left(\frac{2}{\sqrt{14}}, \frac{3-i}{\sqrt{14}} \right)^T$

$\lambda_2 = 1$ has a unit eigenvector $\left(\frac{-5}{\sqrt{35}}, \frac{3-i}{\sqrt{25}} \right)^T$

$$Q = \begin{pmatrix} \frac{2}{\sqrt{14}} & -\frac{5}{\sqrt{35}} \\ \frac{3-i}{\sqrt{14}} & \frac{3-i}{\sqrt{35}} \end{pmatrix}$$

(c) $\lambda_1 = 3$ has a unit eigenvector $\left(-\frac{1}{\sqrt{2}}, \frac{i}{\sqrt{2}}, 0 \right)^T$

$\lambda_2 = 2$ has a unit eigenvector $(0, 0, 1)^T$

$\lambda_3 = 1$ has a unit eigenvector $\left(\frac{1}{\sqrt{2}}, \frac{i}{\sqrt{2}}, 0 \right)^T$

$$Q = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 & 0 & 1 \\ i & 0 & i \\ 0 & \sqrt{2} & 0 \end{pmatrix}$$

2. Consider $A = \begin{pmatrix} 16a-12 & 12a-9 \\ 12a+16 & 9a+12 \end{pmatrix}$.

(a) Show that $\det(A) = 0$ and find an eigenvector of A corresponding to eigenvalue 0.

(b) Suppose the algebraic multiplicity of 0 > geometric multiplicity of 0. Show that $a = 0$. Find an orthogonal matrix Q and an upper triangular matrix T such that $A = QTQ^T$.

Solution

(a) $\det \begin{pmatrix} 16a-12 & 12a-9 \\ 12a+16 & 9a+12 \end{pmatrix} = (16a-12)(9a+12) - (12a-9)(12a+16)$

$= 4(4a-3)3(3a+4) - 3(4a-3)4(3a+4) = 0$

$N \begin{pmatrix} 16a-12 & 12a-9 \\ 12a+16 & 9a+12 \end{pmatrix} = \text{span} \left\{ \begin{pmatrix} 3 \\ -4 \end{pmatrix} \right\}$

$\begin{pmatrix} 3 \\ -4 \end{pmatrix}$ is an eigenvector of A corresponding to eigenvalue 0.

(b) If $a \neq 0$, then A has 2 distinct eigenvalues. By Corollary 6.2.4, A is diagonalizable. By Theorem 6.2.3, algebraic multiplicity of 0 = geometric multiplicity of 0. Contradiction.

Clearly, $\begin{pmatrix} 4 \\ 3 \end{pmatrix}$ is orthogonal to $\begin{pmatrix} 3 \\ -4 \end{pmatrix}$. Choose $Q = \begin{pmatrix} 0.6 & 0.8 \\ -0.8 & 0.6 \end{pmatrix}$, an orthogonal matrix.

$$T = Q^T AQ = \begin{pmatrix} 0.6 & -0.8 \\ 0.8 & 0.6 \end{pmatrix} \begin{pmatrix} -12 & -9 \\ 16 & 12 \end{pmatrix} \begin{pmatrix} 0.6 & 0.8 \\ -0.8 & 0.6 \end{pmatrix} = \begin{pmatrix} 0.6 & -0.8 \\ 0.8 & 0.6 \end{pmatrix} \begin{pmatrix} 0 & -15 \\ 0 & 20 \end{pmatrix} = \begin{pmatrix} 0 & -25 \\ 0 & 0 \end{pmatrix}.$$

3 Consider $-3x^2 + 6xy + 5y^2 - 24 = 0$.

(i) Find a suitable change of coordinates (i.e., a rotation and/or a translation) so that the resulting conic section is in standard form,

(ii) Identify the curve.

Solution

$$A = \begin{pmatrix} -3 & 3 \\ 3 & 5 \end{pmatrix}.$$

The eigenvalues are $\lambda_1 = 6$, $\lambda_2 = -4$ with orthonormal eigenvectors

$$\left(\frac{1}{\sqrt{10}}, \frac{3}{\sqrt{10}}\right)^T \quad \text{and} \quad \left(-\frac{3}{\sqrt{10}}, \frac{1}{\sqrt{10}}\right)^T, \quad \text{respectively.}$$

Let

$$Q = \frac{1}{\sqrt{10}} \begin{pmatrix} 1 & -3 \\ 3 & 1 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} x' \\ y' \end{pmatrix} = Q^T \begin{pmatrix} x \\ y \end{pmatrix}$$

The equation simplifies to

$$6(x')^2 - 4(y')^2 = 24$$

$$\frac{(x')^2}{4} - \frac{(y')^2}{6} = 1$$

The graph will be a hyperbola.

4. Let A be a 2×2 matrix such that $A = UTU^H$ with unitary matrix U and upper triangular matrix T. Suppose $t_{12} \neq 0$. Show that

(a) the eigenvalues of A are $\lambda_1 = t_{11}$ and $\lambda_2 = t_{22}$.

(b) $\boldsymbol{u}_1$, the first column of U, is an eigenvector of A belonging to $\lambda_1 = t_{11}$.

(c) $\boldsymbol{u}_2$, the second column of U, is not an eigenvector of A belonging to $\lambda_2 = t_{22}$.

Solution

(a) $p_A(\lambda) = \det(A - \lambda I)$

$= \det(UTU^H - \lambda I)$

$= \det(UTU^{-1} - U\lambda IU^{-1})$

$= \det U \det(T - \lambda I) \det U^{-1}$

$= \det(T - \lambda I)$

$= (t_{11} - \lambda)(t_{22} - \lambda)$

(b) $T = \begin{pmatrix} t_{11} & t_{12} \\ 0 & t_{22} \end{pmatrix}, U = \begin{pmatrix} \boldsymbol{u}_1 & \vdots & \boldsymbol{u}_2 \end{pmatrix}, U^H = U^{-1}$

$A = UTU^H \quad \Rightarrow \quad AU = UT$

$A\begin{pmatrix} \boldsymbol{u}_1 & \vdots & \boldsymbol{u}_2 \end{pmatrix} = \begin{pmatrix} \boldsymbol{u}_1 & \vdots & \boldsymbol{u}_2 \end{pmatrix}\begin{pmatrix} t_{11} & t_{12} \\ 0 & t_{22} \end{pmatrix}$

$A\boldsymbol{u}_1 = t_{11}\boldsymbol{u}_1$

5. Show that if a matrix U is both unitary and Hermitian, then any eigenvalue of U must equal either 1 or -1.

Solution

Let $\boldsymbol{v}$ be an eigenvector of A corresponding to eigenvalue λ. Since U is unitary,

$$\|U\boldsymbol{v}\|^2 = \|\lambda\boldsymbol{v}\|^2$$
$$\boldsymbol{v}^H U^H U\boldsymbol{v} = (\lambda\bar{\lambda})\boldsymbol{v}^H\boldsymbol{v}$$
$$\boldsymbol{v}^H\boldsymbol{v} = (\lambda\bar{\lambda})\boldsymbol{v}^H\boldsymbol{v}$$
$$1 = |\lambda|$$
$$\lambda = \cos\theta + i\sin\theta$$

Since U is Hermitian, λ must be real. $|\lambda| = 1 \quad \Rightarrow \quad \lambda = \pm 1$.